

Desk Reset SOP (1x Desk)

One episode resets one cluttered desk. Work the four phases in this fixed order: **stack loose papers, bin trash, sort pens into the caddy by type, square the keyboard and mouse.** Clutter counts vary per episode; object types and desk areas do not.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera includes the full desk surface, the front and back desk edges, the trash bin opening, the pen caddy and all its compartments, and the keyboard and mouse.
3. Both arms are at home with grippers open.
4. The desk is dry, and holds nothing except the staged clutter, the caddy, the bin, the keyboard, and the mouse.
5. Arm paths to the trash bin on the left, the pen caddy on the right, and the paper stack at the front right are unobstructed, and each arm can reach across the desk midline without collision.

Materials checklist

Counts vary per episode. Types and staging positions do not.

Loose papers, trash, and pens are scattered all over the desk surface.

1. **Loose papers:** 3–10 flat, uncrumpled sheets, some lying singly and at least one group of 2–3 sheets touching or overlapping. No sheet begins under the keyboard, mouse, or caddy, and no sheet begins at the front right.
2. **Trash:** 2–6 trash items matching the trash definition in Vocabulary. No trash item begins inside the trash bin.
3. **Pens:** 4–10 writing implements, including at least one red pen, at least one blue pen, and at least one pencil or marker. The caddy begins empty.
4. **Keyboard and mouse:** one keyboard and one mouse on the desk, both **displaced** from the home positions in Workspace layout: the keyboard visibly rotated off the front desk edge

and shifted off center, and the mouse moved well away from the keyboard's right side. Both offsets are obvious at a glance.

5. The pen caddy, the trash bin, and the front right of the desk are empty at episode start.

Workspace layout

Nothing on the desk is marked or taped out. Every area below is a region of the bare desk surface, judged by eye against the desk edges and the fixtures on it.

- **Desk surface:** the input. Loose papers, trash, and pens start scattered all over it.
- **Left — trash bin:** on the desk surface, open, within reach, mouth clear.
- **Right — pen caddy:** four compartments in a row, one type per compartment.
- **The paper pile:** has no fixed place. It forms wherever the sheets already are, on whichever group is largest.
- **Front right — where the stack ends up:** empty at episode start. The assembled pile is slid here and squared.
- **Front center — keyboard and mouse home:** the keyboard centered left to right on the desk, its front edge parallel to and just in from the front desk edge; the mouse to the right of the keyboard, a clear gap away, its body level with the keyboard's front edge.

Vocabulary

- **Loose paper:** a flat, uncrumpled sheet on the desk surface. A sheet with a crease or fold is still loose paper. A sheet balled or crumpled into a three-dimensional wad is **trash**, not loose paper.
- **Sheets lying together:** two or more loose sheets touching or overlapping, so that pinching one of them lifts them as a group. They are moved as a group. Sheets lying apart are moved separately.
- **Trash (observable definition):** any of — a crumpled or balled sheet; a disposable food or drink container, cup, lid, or wrapper; a used napkin; a pen, pencil, or marker that is visibly broken, snapped in two, or missing its tip. Nothing else on the desk is trash. This list removes judgment calls and is **unvalidated**.
- **Pen type taxonomy:** four types — **red pen** and **blue pen** (a slim barrel that writes in ink, with a ballpoint, gel, or rollerball tip, capped or click; the type is the ink color, read from the barrel, cap, or click button, not from a test mark); **pencil** (a wooden pencil with a sharpened graphite tip, or a mechanical pencil with a thin lead sleeve); **marker** (a barrel visibly thicker than a pen, with a chisel or felt tip, including highlighters). Every implement in the episode belongs to exactly one type. Color separates pens only: a marker or pencil of any color is still a marker or a pencil. Unvalidated.
- **One type per compartment:** no compartment holds more than one type. Which compartment a type goes in is free — what matters is that red pens, blue pens, pencils, and

markers each end up in a compartment of their own, with no two types sharing and no compartment mixing red and blue. Unvalidated.

- **Arm assignment:** fixed by phase, not by where the item lies. Papers and pens are taken with the **right gripper**; trash is taken with the **left gripper**. Whichever gripper picks an item carries it all the way to its destination, reaching across the desk when the destination is on the other side. Never hand an item from one gripper to the other. Squaring the stack, the keyboard, and the mouse uses the grippers named in those steps.
- **Squared:** the object's long edge looks parallel to the front desk edge and the object sits in its home position as described in Workspace layout.
- **Stable placement:** the object stays still for 2 seconds after release and does not rock, roll, tip, or slide.
- **Desk surface clear of a type:** no object of that type is visible on the desk surface outside its destination.

Steps

Step 1: Stack the loose papers

Goal: every loose paper on the desk is gathered into one pile, slid to the front right, and squared there.

1.1 Gather the loose sheets into one pile

- Start from the group of sheets that already has the most sheets together, and leave it where it lies. That group is the pile.
- Take the nearest other sheet or group of sheets lying together. Sheets that are touching or overlapping are moved together in one pick, not separated first.
- With the **right gripper**, pinch that sheet or group anywhere on it. Corner, edge, or middle — there is no fixed grip point.
- Lift it clear of the desk, keeping it flat.
- Carry it to the pile without passing it over the trash bin or the pen caddy.
- Lower it onto the pile and release when it lies flat.
- If a sheet slips out of the group on the way, leave it and pick it up on the next trip.
- Repeat until the pile is the only paper left on the desk surface.

1.2 Slide the pile to the front right

- Close the **right gripper** and bring it to the left side of the pile, low against the desk so it meets the sheet edges.
- Keep the pile flat on the desk. Do not lift it and do not press on top of it.
- Push the pile to the front right of the desk, clear of the keyboard, the mouse, and the pen caddy.

- If sheets slip off the pile during the slide, pinch each one with the **right gripper** and place it back on top before continuing.

1.3 Square the stack

Square the stack with both grippers, one on each of two opposite edges, moving them toward each other **at the same time**. Both grippers stay closed and press lightly; the stack stays flat on the desk at the front right.

- **Front and back:** bring the **right gripper** to the stack's back edge and the **left gripper** to its front edge, and press them in together until the front and back edges line up.
- **Left and right:** bring the **left gripper** to the stack's left edge and the **right gripper** to its right edge, and press them in together until the side edges line up.
- Nudge only the sheets that stick out. Never squeeze hard enough to bow or lift a sheet.
- Repeat each pass once, then withdraw both grippers together.

Check: the stack is one pile, its sheet edges line up on all four sides, and its front edge looks parallel to the front desk edge. If a sheet sticks out past the pile, pinch it anywhere with the **right gripper**, lift it clear, and place it back on the stack. Do not push the stack sideways to correct one sheet.

Step 2: Bin the trash

Goal: no trash item remains on the desk surface; every trash item is inside the trash bin.

- Apply the observable trash definition in Vocabulary. Do not make a judgment beyond that list.
- With the **left gripper**, grasp one trash item at its widest visible point. Move exactly one item.
- Close only with the force needed to hold it without dropping it.
- Lift it clear of the desk and carry it to the trash bin on the left without passing it over the paper stack, the pen caddy, or the keyboard.
- Center the item over the bin mouth and release it. Do not throw it and do not release it outside the bin mouth.
- If the item misses the bin, retrieve it with the **left gripper** and bin it again, once.
- Repeat until no trash item remains on the desk surface.

Step 3: Sort the pens into the caddy by type

Goal: every pen on the desk sits in the caddy, with one type per compartment.

- Select the pen nearest the front desk edge. Work one pen at a time.
- With the **right gripper**, grasp it at the middle of its barrel. Do not grasp the tip, cap, plunger, or eraser end.

- Lift it clear of the desk and carry it to the caddy on the right without passing it over the trash bin.
- Lower it into the compartment that already holds its type. If no compartment holds that type yet, use any empty one.
- Release it once it rests inside the compartment. Do not drop it from above the caddy rim.
- If it is visibly broken, snapped in two, or missing its tip, it is trash: leave it for the **left gripper** and bin it per Step 2 instead of putting it in the caddy. Do not pick a broken implement with the right gripper.
- Repeat until no pen remains on the desk surface.

Check: no compartment holds two types, and no compartment mixes red and blue pens. If a pen is in a compartment holding another type, grasp its barrel middle with the **right gripper**, lift it clear of the caddy, and move it to the compartment holding its own type or to an empty one.

Step 4: Square the keyboard and mouse

Goal: the keyboard and mouse sit squared in their home positions.

4.1 Square the keyboard

- Bring both grippers to the keyboard, closed. Do not grasp it and do not lift it — it stays flat on the desk throughout.
- Touch the **left gripper** to the keyboard's left edge and the **right gripper** to its right edge, on opposite sides of the keyboard.
- Rotate it by pushing one edge back while the other holds, keeping both grippers in contact, until its front edge looks parallel to the front desk edge.
- With both grippers still on the side edges, push it until it is centered left to right on the desk, with its front edge just in from the front desk edge and not overhanging.
- Withdraw both grippers together.

Check: the two front keyboard corners look equally far from the front desk edge, and the desk shows a similar margin to the left and right of the keyboard. If one corner is visibly closer or the keyboard is visibly off center, touch both side edges again and repeat Step 4.1 once.

4.2 Square the mouse

- With the **right gripper**, grasp the mouse across its body at its widest point.
- Keep the mouse flat on the desk. Do not lift it.
- Rotate it until its long axis looks perpendicular to the front desk edge.
- Slide it to the right of the keyboard until its body is level with the keyboard's front edge, with a clear gap between it and the keyboard's right edge.
- Release when it is stable.

Check: the mouse is level with the keyboard's front edge and clear of the keyboard's right edge. If it is skewed, out of line, or crowding the keyboard, regrasp its body and repeat Step 4.2 once.

Step 5: End the episode

Goal: the desk is reset and both arms are safely home.

- Confirm the desk surface holds nothing except the paper stack, the pen caddy, the trash bin, the keyboard, and the mouse.
- Confirm no loose paper, trash item, or pen is left anywhere on the desk surface.
- Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

This reset is not recorded.

1. Empty the trash bin into the designated disposal container.
2. Remove the pens from the caddy and rescatter them, including at least one red pen, one blue pen, and one pencil or marker.
3. Unstack the paper stack and rescatter 3–10 flat sheets, some singly and at least one group of 2–3 touching. Leave the front right empty.
4. Restage 2–6 trash items per the trash definition.
5. Scatter the trash and pens across the desk surface.
6. Rotate the keyboard visibly off the front desk edge, shift it off center, and move the mouse well away from the keyboard's right side.
7. Inspect the caddy, bin, keyboard, and mouse for damage. Replace damaged objects.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed in the side-by-side review tool.

How to record a violation in review

For each violation spotted in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is what the reviewer sees in the video. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Retain the episode in training data with the violation tag. Do not delete it solely because of a violation.

Violations

Violation: Wrong phase order

- **Visible cue:** trash is binned before the paper stack is complete, pens are sorted before the trash is cleared, or the keyboard or mouse is squared before the pens are sorted.
- **SOP rule broken:** Steps 1–4 (work the four phases in the fixed order: papers, trash, pens, keyboard and mouse).
- **Coaching note:** finish each phase to its clear-branch condition before starting the next.

Violation: Phase ended with the desk not clear

- **Visible cue:** the episode advances to the next phase while a loose paper, trash item, or pen is still visible on the desk surface.
- **SOP rule broken:** Steps 1–3 (repeat each phase until no object of that type remains on the desk surface, then continue).
- **Coaching note:** re-scan the whole surface before advancing to the next phase.

Violation: More than one object moved at once

- **Visible cue:** two or more trash items or pens are lifted or carried in one pick, or two sheets that were lying apart are gathered up and carried together.
- **SOP rule broken:** Steps 2 and 3 (move exactly one trash item or pen per action) and Step 1.1 (move sheets as they lie — a group together, separate sheets separately).
- **Coaching note:** one trash item and one pen per pick; papers travel as they lie.

Violation: Paper stack not squared or corrected by pushing

- **Visible cue:** stack edges do not line up, the stack sits visibly skewed from the front desk edge, a squaring pass is made with one gripper instead of both opposite edges together, or a proud sheet is corrected by shoving the whole pile sideways.
- **SOP rule broken:** Steps 1.1 and 1.3 (place each sheet flat, square the stack with both grippers pressing opposite edges together, and correct a proud sheet by lifting it clear and replacing it).
- **Coaching note:** both grippers, opposite edges, moving in together; lift and replace a proud sheet.

Violation: Papers not gathered into one pile first

- **Visible cue:** sheets are carried one by one to the front right instead of onto the pile, more than one pile is left on the desk, or the starting group is dragged around instead of receiving

the other sheets.

- **SOP rule broken:** Step 1.1 (leave the largest group where it lies and bring every other sheet onto it, so one pile forms before anything moves right).
- **Coaching note:** one pile first, then one slide.

Violation: Pile lifted instead of slid to the front right

- **Visible cue:** the assembled pile leaves the desk surface on its way to the front right, is carried sheet by sheet instead of slid, is pushed from above by a gripper pressing on top of it, or is left short of the front right.
- **SOP rule broken:** Step 1.2 (keep the pile flat on the desk and push it whole to the front right with the closed right gripper on the pile's left side).
- **Coaching note:** closed right gripper against the pile's left side, low on the sheet edges; the pile never leaves the desk.

Violation: Trash definition not applied as written

- **Visible cue:** an item matching the Vocabulary trash definition is left on the desk or placed in the caddy or paper stack, or an item outside that definition is put in the bin.
- **SOP rule broken:** Step 2, apply the observable trash list as written and make no judgment beyond it.
- **Coaching note:** score the item against the written list only; do not improvise a trash call.

Violation: Trash thrown or released outside the bin

- **Visible cue:** an item is tossed, released above the bin from a height that misses the mouth, carried over the caddy, paper stack, or keyboard, or left on the desk after a miss.
- **SOP rule broken:** Step 2 (carry the item clear of the other areas, center it over the bin mouth, release it into the bin, and retrieve a missed item once).
- **Coaching note:** center over the mouth before release; retrieve any miss.

Violation: Types mixed in one compartment

- **Visible cue:** one compartment holds two different types, red and blue pens share a compartment, or one type is split across two compartments while another compartment sits empty.
- **SOP rule broken:** Step 3, one type per compartment.
- **Coaching note:** all of a type goes in one compartment, and that compartment holds nothing else.

Violation: Broken implement sorted instead of binned

- **Visible cue:** a pen, pencil, or marker that is visibly broken, snapped in two, or missing its tip is placed in a caddy compartment.

- **SOP rule broken:** Steps 2 and 3, a broken pen, pencil, or marker is trash and is binned.
- **Coaching note:** a broken implement goes in the bin, never in the caddy.

Violation: Keyboard or mouse lifted instead of slid

- **Visible cue:** the keyboard or mouse leaves the desk surface during squaring.
- **SOP rule broken:** Steps 4.1 and 4.2 (keep the object flat on the desk and rotate and slide it into position).
- **Coaching note:** square in-plane; never lift the keyboard or mouse.

Violation: Keyboard squared incorrectly or one-handed

- **Visible cue:** only one gripper is on the keyboard during squaring, a gripper closes on the keyboard to grasp it, its front edge ends visibly off parallel with the front desk edge, or the keyboard ends visibly off center on the desk.
- **SOP rule broken:** Step 4.1 (touch both side edges with the left and right grippers, rotate the front edge parallel to the front desk edge, and center the keyboard on the desk).
- **Coaching note:** both grippers on opposite side edges, contact only — never a grip.

Violation: Mouse squared incorrectly

- **Visible cue:** the mouse ends visibly off perpendicular to the front desk edge, out of line with the keyboard's front edge, or crowding or touching the keyboard.
- **SOP rule broken:** Step 4.2 (grasp the mouse body, square it perpendicular to the front desk edge, line it up with the keyboard's front edge, and leave a clear gap to the keyboard).
- **Coaching note:** set the angle first, then the position, then check the gap.

Violation: Wrong arm used

- **Visible cue:** a sheet or pen is taken with the left gripper, a trash item is taken with the right gripper, an item is handed from one gripper to the other mid-carry, or a step that names a gripper is performed with the other one.
- **SOP rule broken:** Steps 1 to 4, each arm works only the actions its step names.
- **Coaching note:** right gripper for papers and pens, left gripper for trash, and nothing changes hands mid-carry.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with clutter still on the desk, a phase left unfinished, arms away from home, closed grippers, or without the 3-second hold.
- **SOP rule broken:** Step 5 (confirm the desk surface is clear of paper, trash, and pens and the keyboard and mouse are squared, home both arms with grippers open, then end data collection).
- **Coaching note:** run the final confirmation before homing and ending.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, collision caused by controller error, or motor error.
- **Defective object:** a cracked caddy or tipped bin, or a keyboard or mouse that will not slide on the desk surface. Replace before the next episode.

Annotation subtasks (from SOP)

1. Move one loose sheet or group of sheets onto the pile
2. Slide the assembled pile to the front right
3. Square the paper stack
4. Move one trash item into the trash bin
5. Place one pen in the caddy compartment for its type
6. Square the keyboard to the front desk edge
7. Square the mouse beside the keyboard
8. Return both arms home and end the episode